



Project 1

Enhancing the Zero-Dimensional Pulsatile Circulation Model
Due Date: 30th September 11:59PM

Introduction

In this assignment, you will explore and enhance a simplified cardiovascular circulation model described in Peskin et al. Ch.1.12 (see syllabus). The original model assumes a fixed heart rate, which leads to non-physiological behavior during exercise. Your task is to modify the model by implementing a feedback controller that adjusts the heart rate based on systemic arterial pressure P_{sa} . You'll simulate exercise, demonstrate its effects on blood pressure and cardiac output, and then improve the model to simulate realistic physiological responses. You should submit your assignment as a pdf in the format of a lab report with a description of your methods, discussion of your results, and an appendix for your code. No use of large-language models are allowed for the completion of this assignment (I will know). The base code is already available to you and the goal of this assignment is to learn how to modify a given model to make enhancements that allow you to solve a specific problem.

Problem 1: Understanding the Base Circulation Model

Description:

The base model simulates the circulation system using parameters like systemic resistance R_s , pulmonary resistance, compliances, and pressures. One key feature is that the heart rate remains fixed throughout the simulation, which does not reflect what happens in the human body during exercise.

- **Run the Simulation:** Run the base model code provided in the original script.
- **Observe and Analyze:** Track the output for systemic arterial pressure P_{sa} and systemic flow $Q_{sa,sv}$.

Problem 2: Limitations of a Fixed Heart Rate during Exercise

Description:

One limitation of the base model is the assumption of a constant heart rate. In reality, during exercise, the heart rate increases to compensate for the drop in systemic resistance R_s .

- **Exercise Simulation:** Lower R_s to simulate exercise (e.g., by halving its value) and observe the effect on P_{sa} and $Q_{sa,sv}$.
- **Results:** Show that in the fixed heart rate model, systemic arterial pressure P_{sa} and cardiac output $Q_{sa,sv}$ decrease significantly, which does not reflect physiological reality.

Problem 3: Implementing a Feedback Controller for Heart Rate

Description:

To better simulate the physiological response, we will implement a feedback controller that adjusts the heart rate based on systemic arterial pressure P_{sa} . Since P_{sa} is highly pulsatile, we need to filter it using a low-pass filter before applying it to the controller. A simple low-pass filter can be implemented by solving the ODE:

$$\frac{dP_{\text{filtered}}}{dt} = \frac{1}{\tau_P} (P_{sa} - P_{\text{filtered}})$$

where τ_P is the filter time constant.

- **Task 1:** Implement this filtering in the code using the Forward Euler method. Insert it into the appropriate location in the simulation to ensure the feedback controller works properly.

Problem 4: Warped Time Execution and Compliance Function

Once the filtered systemic arterial pressure P_{filtered} is computed, the feedback controller will detect when P_{filtered} drops below a set pressure P^* , which represents the desired mean arterial pressure, and increase the heart rate accordingly. The heart rate function $F(t)$ can be updated as:

$$F(t) = F_{\text{heart}} + \Delta F$$

where

$$\Delta F = c F_{\text{heart}} \frac{P_{\text{filtered}} - P^*}{P^*}$$

Here, c is the gain of the feedback system (negative), F_{heart} is the steady heart rate, P_{filtered} is the filtered P_{sa} , and P^* is the set pressure.

- **Task 1:** Implement this feedback controller in the code to adjust the heart rate dynamically during the simulation.

Description:

In order to implement the time-varying heart rate, the compliance function must be updated correctly. The compliance function now depends on a “warped” time $t_{\text{heart}}(t)$, which is determined by the time-varying heart rate. The warped time $t_{\text{heart}}(t)$ allows the compliance function to reflect changes in heart rate, simulating physiological behavior during exercise.

To implement $F(t)$ into our compliance function which drives our model, we consider our known periodic function for compliance, $C_{\text{heart}}(t_{\text{heart}})$ with period T_{heart} and heart rate $F_{\text{heart}} = 1/T_{\text{heart}}$. To produce continuously varying heartrate $F(t)$, we set our new compliance function to look like:

$$C(t) = C_{\text{heart}}(t_{\text{heart}}(t)) \quad (1)$$

where the warped time can be updated by solving the following ODE:

$$\frac{dt_{\text{heart}}}{dt} = \frac{F(t)}{F_{\text{heart}}}$$

where $F(t)$ is the time-varying heart rate, and F_{heart} is the steady heart rate. This equation warps the time execution of the compliance function, allowing for varying heart rates.

- **Task:** Update how the compliance function is called in the code by solving this ODE using the Forward Euler method. Modify the time execution of the compliance function to account for the varying heart rate. Make sure to do this for both ventricles!

Problem 5: Running the Full Simulation with Exercise

Description:

Now, run the full simulation with the feedback-controlled heart rate. Allow the simulation to reach a periodic steady state, then simulate the start of exercise by lowering R_s at a specific time. After reaching a new steady state, simulate the end of exercise by restoring R_s . Ensure that the system reaches a periodic steady state after exercise stops. You can adjust the duration of the full simulation by changing the `klokmax` parameter

- **Task:** Run the simulation and report how the system responds during and after exercise. Discuss the improvements in P_{sa} and $Q_{\text{sa,sv}}$. Also plot $F(t)$.

Problem 6: Improving the Model with a Dynamic Set Pressure

Description:

While the model is improved by introducing a feedback controller, you may notice that the set pressure P^* remains constant throughout the simulation. However, in reality, our bodies anticipate exercise by raising P^* at the onset of exercise and lowering it back to normal just after exercise ends.

- **Task:** Implement this dynamic set pressure in the simulation. Increase P^* at the start of exercise and return it to normal just after the end of exercise. Compare what happens when you return the set pressure back to the mean arterial pressure right at the end of exercise vs. slightly after. Report how this change in set pressure improves the behavior of the model. You may look into literature values for heart rate and blood pressure during exercise to calibrate your model to a physiologically reasonable range (by adjusting the gain of your feedback controller).